Songyao Xue

Fall 2027 Research Applicant

Luoyang, Henan, China 18637229290@139.com

Education

Luoyang Normal University — B.Eng. in Software Engineering Sep. 2023 – Jul. 2027 (expected)

- GPA: 3.53/5.00; academic ranking: Top 10%.
- Relevant coursework: Algorithm Design and Analysis, Data Structures, Discrete Mathematics, Linear Algebra, Probability and Statistics, Operating Systems, Computer Networks, Database Systems, and programming in C/Java/Python/JavaScript.
- English: CET-4 477; CET-6 first attempt 414. Mandarin Proficiency: Level 2-A.

Research Interests

Quantum cryptography and post-quantum security; quantum signatures and protocol composition; simulation-based security and secure multi-party computation; quantum information and algorithms.

Research Experience

Quantum Information, Cryptography, Communication, and Game Theory 2023–Present

Undergraduate Research Assistant, Luoyang Normal University

- Conduct literature review, mathematical modeling, theoretical derivation, and simulation-based validation across quantum cryptography, communication, game theory, and probabilistic decoding.
- Use Python, Qiskit, PennyLane, numerical experiments, graph models, optimization, and semidefinite-program certificate logic to test protocol behavior, noise robustness, and explicit resource trade-offs.
- Prepare English LaTeX manuscripts, appendix proofs, figures, tables, notation, and bibliographies through reproducible research workflows.

Supervisor-led Reading Group in Quantum Information and Communication 2023–Present

Research Trainee, Luoyang Normal University

- Participate in weekly seminars on quantum states and circuits, algorithms, quantum communication, quantum key distribution, cryptography, protocol security, and mathematical modeling.
- Critically review papers and reinforce theory through Qiskit/PennyLane exercises and numerical simulation.

Manuscripts and Research Projects

On Grover-Based Quantum Signature Schemes 2025–2026

Second author; submitted to Entropy; revised and resubmitted

- Contributed to analysis of key-recovery, forgery, and repudiation pathways arising from verification structure and information flow; helped abstract four reusable protocol-inspection points.
- Supported mathematical framework design, a strengthened arbitrated construction, Qiskit verification, manuscript writing, LaTeX preparation, and reference management.

A Learnable Unified Framework for Weak-Link Multi-Group Quantum Games 2025–2026

First author; submitted to Quantum Machine Intelligence; under editorial review

- Proposed a signed weighted-graph model for cooperation, competition, and weak coupling; derived game tensors, first-order advantage conditions, task-dependent sign rules, and spectral-radius constraints.
- Implemented automated search, repeated trials, topology/noise comparisons, training diagnostics, and backend checks while separating theorem-level, surrogate, and proxy quantities.

Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding 2025–2026

First author; manuscript in preparation; planned submission in 2026

- Constructed grouped-ancilla, conditional-discrimination, QEC, and MAP-decoding methods with explicit resource-normalized criteria.
- Developed analytical sufficient conditions, small-instance SDP certificate logic, Qiskit/Python parameter scans, and resource phase diagrams.

Weak-Link Entanglement-Assisted No-Signaling Quantum Communication Protocols 2026

First author; manuscript in preparation; planned submission in 2026

- Developed a task-aware communication model using multi-group entanglement, weak-link correlations, no-signaling constraints, and unified noise-to-visibility mappings.

Selected Technical Projects

Reflectance-Based Epitaxial Layer Thickness Estimation Sep. 2025

Mathematical modeling project; core developer and team leader

- Built optical/physical models using FFT and the Transfer Matrix Method; implemented the Python pipeline and Bootstrap confidence intervals.

Archimedean Spiral-Based Optimization Design Model Sep. 2024

Mathematical modeling project; core developer and team leader

- Modeled parametric trajectories and applied particle swarm optimization; contributed to modeling, code, and the written report.

Technical Skills

Theory/Methods	Quantum protocol and information-flow analysis; graph modeling; first-order and spectral analysis; optimization; POVM-based state discrimination; semidefinite programming; Bayesian/MAP decisions.
Computing	Python, NumPy, Pandas, Matplotlib, Qiskit, PennyLane, numerical simulation, noise/sensitivity analysis; C, Java, JavaScript, SQL.
Research	LaTeX, Overleaf, TeXstudio, English manuscript preparation, reference management, reproducible experiments, Git/Linux/Ubuntu.

Honors and Service

- Provincial Third Prize, Chinese Mathematics Competition, Henan Province (2024, 2025).
- Provincial Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Henan Province (2025).
- Honorable Mention, Shuwei Cup Mathematical Modeling Challenge (2026).
- Outstanding Student Cadre and Merit Student Award, Luoyang Normal University (2025).
- Chief Administrator, Mathematical Modeling Laboratory (2023–Present): organize discussions, meetings, training, and project-based learning.

Research Availability

Senior-year coursework will be light; available for substantial, potentially near-full-time research from Fall 2026 through Summer 2027 before Fall 2027 entry.

Personal and Research Statement

Songyao Xue
Luoyang Normal University | Fall 2027

Research Direction

My undergraduate research has taught me to distrust a convenient shortcut: that a system is secure or useful merely because its components are powerful. A quantum signature protocol may invoke legitimate quantum primitives and still expose verification structure. A decoder may improve reliability but lose its advantage after ancillary and error-correction costs are counted. A weak entanglement link may appear defective until its role is aligned with a task. Across these problems, I have learned to ask what assumptions, interfaces, and resource roles actually justify a claim.

I study Software Engineering at Luoyang Normal University, where I hold a GPA of 3.53/5.00 and rank in the top 10%. Since 2023 I have participated in a supervisor-led reading group and research programme in quantum information, communication, cryptography, and game theory. Software Engineering gave me a practical habit of decomposing systems into interfaces and testable components. Research has gradually pushed me from implementing known models toward writing mathematical abstractions, identifying structural failures, and maintaining reproducible evidence.

Protocol Security

My closest preparation for theoretical cryptography is collaborative work on Grover-based quantum signature schemes. The problem was not whether Grover operations, QKD, QOTP, or teleportation are individually legitimate. It was whether their composition protected keys, message origin, and verification authority against realistic participants and communication access.

I helped organize the mathematical framework and notation, supported four reusable inspection points, implemented Qiskit checks, and contributed to a strengthened arbitrated construction and manuscript preparation. Our analysis examined key-recovery, forgery, and repudiation pathways produced by public verification structure and information flow. I am the second author; the manuscript has been revised and resubmitted.

The project gave me an important but incomplete foundation. I learned to trace a protocol and state an attack, but I have not yet mastered the formal machinery that turns security intuition into a reduction or simulation theorem. In particular, I need deeper training in complexity theory, modern cryptographic definitions, zero knowledge, secure computation, non-malleability, extraction, concurrency, and post-quantum proof techniques. I want my next stage of research to close precisely this gap.

Analytical Models and Evidence

My first-author projects developed complementary research habits. In weak-link multi-group quantum games, I represented groups as vertices and heterogeneous resources as signed weighted edges. I derived game tensors, sign rules, first-order conditions, and spectral constraints, then built automated search, repeated-trial, noise, and backend checks. The manuscript is under editorial review. The work trained me to distinguish theorem-level results from surrogate objectives and backend proxy quantities.

In probabilistic quantum decoding, I combined source-shaped state design, grouped ancilla, conditional POVMs, QEC reliability, MAP decisions, and resource-normalized criteria. I developed analytical sufficient conditions, small-instance SDP certificate logic, parameter sweeps, and sensitivity analysis. Representative negative regimes were as important as favorable ones: they showed when an apparent reliability gain disappeared after overhead was counted. A further weak-link communication manuscript studies how heterogeneous entanglement can be mapped through common visibility and task models. Both manuscripts remain in preparation.

These projects are not cryptographic security results, and I would not present them as such. Their value for my next step is methodological: stating restricted claims, making assumptions visible, checking analytical structure with computation, and keeping negative evidence. In cryptography, the relevant accounting will shift from physical overhead toward adversary models, round complexity, assumptions, simulator access, and reduction loss.

Why CryptoPlus and What I Would Do First

CryptoPlus is unusually explicit that its research is mathematical and theoretical, with cryptography connected to complexity, algorithms, quantum computing, and information theory. That description is important to me because it identifies both my strongest bridge and my largest gap. Xiao Liang's recent work on post-quantum and fully quantum simulation, extraction, non-malleability, and secure computation addresses the exact level of rigor that my protocol-security experience now needs.

I would begin modestly. First, I would complete a structured reading and proof-writing programme in modern cryptography, complexity, and quantum information. Second, I would reconstruct a compact result or counterexample from the post-quantum signature, commitment, or simulation literature and produce an explicit map of definitions, assumptions, rounds, and adversary access. Third, under supervision, I would formalize a small signature or commitment case study: write the security game, trace composition assumptions, and test one bounded change such as concurrency or quantum access. Python or Qiskit would support small sanity checks, never substitute for proof.

I can contribute immediately through careful reading notes, formal notation, reproducible checks, bibliography and LaTeX discipline, and persistence with negative or incomplete results. From Fall 2026, my senior-year course load will be light, so I can devote substantial and potentially near-full-time effort through Summer 2027. An internship before formal entry would allow both the group and me to evaluate research fit through actual work.

Long-Term Objective

My long-term interest remains reliable quantum information processing, but my intended contribution is becoming more precise: I want to understand when security claims survive quantum adversaries, composition, and concurrency. Quantum cryptography is the most rigorous setting in which to pursue that question because every interface and assumption must be made explicit.

I am applying with a research-first priority. The most useful path would be an internship beginning in Fall 2026 and, if the research fit develops, Fall 2027 PhD study. I would also consider another route recommended by the supervisor; I understand that the current MPhil route is self-funded. I do not expect my Software Engineering background or simulation record to replace theoretical preparation. I expect them to help me build dependable research habits while I acquire the mathematical depth required to become an independent cryptography researcher.

洛阳师范学院学生成绩明细(有效)

院(系)/部：软件学院　　行政班级：2023级软件工程4班　　平均学分绩点：3.53

学　　号：231164193　　姓　　名：薛淞耀　　打印时间：2026-07-16

学年学期：2023-2024学年第一学期

序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
1	[00001214]劳动教育（理论）	1.0	公共课/必修课	考试	初修	83.00	1.0	3.3	3.30	
2	[0000G01]军事理论	2.0	公共课/必修课	考试	初修	95.00	2.0	4.5	9.00	
3	[0015000062]思想道德与法治	3.0	公共课/必修课	考查	初修	87.00	3.0	3.7	11.10	
4	[0016000016]大学体育（一）	1.0	公共课/必修课	考试	初修	78.00	1.0	2.8	2.80	
5	[0017000TY6]大学英语（一）	3.0	公共课/必修课	考试	初修	82.00	3.0	3.2	9.60	
6	[12000015]大学生心理健康教育	2.0	公共课/必修课	考查	初修	77.00	2.0	2.7	5.40	
7	[15000711]形势与政策（一）	0.5	公共课/必修课	考试	初修	74.00	0.5	2.4	1.20	
8	[0008002014]大学计算机基础	1.5	专业课/必修课	考查	初修	82.00	1.5	3.2	4.80	
9	[0008002016]学科专业导论	0.5	专业课/必修课	考查	初修	合格	0.5	1.5	0.75	
10	[0008103020]电子技术	4.5	专业课/必修课	考试	初修	66.00	4.5	1.6	7.20	
11	[0008202002]C语言程序设计	5.0	专业课/必修课	考试	初修	70.00	5.0	2.0	10.00	
12	[0008502011]高等数学一	4.0	专业课/必修课	考试	初修	82.00	4.0	3.2	12.80	
13	[00001216]当代大学生国家安全教育	1.0	公共课/任选课	考试	初修	93.00	1.0	4.3	4.30	

学年学期：2023-2024学年第二学期

序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
1	[0015000016]中国近现代史纲要	3.0	公共课/必修课	考试	初修	66.00	3.0	1.6	4.80	
2	[0016000017]大学体育（二）	1.0	公共课/必修课	考试	初修	83.00	1.0	3.3	3.30	
3	[0017000TY7]大学英语（二）	3.0	公共课/必修课	考试	初修	84.00	3.0	3.4	10.20	

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学　　号：231164193　　姓　　名：薛淞耀　　打印时间：2026-07-16

学年学期：2023-2024学年第二学期

序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
4	[15000712]形势与政策（二）	0.5	公共课/必修课	考试	初修	77.00	0.5	2.7	1.35	
5	[91000001]职业生涯规划与职业素质	1.0	公共课/必修课	考试	初修	98.00	1.0	4.8	4.80	
6	[0008102004]高等数学（二）	4.0	专业基础课/必修课	考试	初修	88.00	4.0	3.8	15.20	
7	[0610Y003]大学物理（1）	2.5	专业基础课/必修课	考查	初修	91.00	2.5	4.1	10.25	
8	[0018103021]面向对象程序设计	5.0	专业课/必修课	考查	初修	82.00	5.0	3.2	16.00	
9	[0008105019]网页设计	3.0	专业课/限选课	考查	初修	81.00	3.0	3.1	9.30	

学年学期：2024-2025学年第一学期

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1	[00007002]创新创业基础	2.0	公共课/必修课	考试	初修	100.00	2.0	5.0	10.00	
2	[0002000006]河洛文化概论	1.0	公共课/必修课	考查	初修	87.00	1.0	3.7	3.70	
3	[0016000018]大学体育（三）	1.0	公共课/必修课	考试	初修	81.00	1.0	3.1	3.10	
4	[0017000TY8]大学英语（三）	1.5	公共课/必修课	考试	初修	83.00	1.5	3.3	4.95	
5	[010000012]大学语文	1.0	公共课/必修课	考查	初修	89.00	1.0	3.9	3.90	
6	[15000021]马克思主义基本原理	3.0	公共课/必修课	考试	初修	78.00	3.0	2.8	8.40	
7	[15000083]形势与政策（三）	0.5	公共课/必修课	考查	初修	91.00	0.5	4.1	2.05	
8	[0008002007]离散数学	4.0	专业基础课/必修课	考查	初修	98.00	4.0	4.8	19.20	
9	[0008102005]线性代数	3.0	专业基础课/必修课	考试	初修	91.00	3.0	4.1	12.30	
10	[0008102008]数据结构	3.5	专业基础课/必修课	考试	初修	86.00	3.5	3.6	12.60	

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序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
11	[0008502041]计算机组成原理	3.5	专业基础课/必修课	考试	初修	79.00	3.5	2.9	10.15	
12	[0610Y004]大学物理（2）	2.5	专业基础课/必修课	考查	初修	92.00	2.5	4.2	10.50	
13	[0008145040]Java课程设计	1.0	专业课/限选课	考查	初修	77.00	1.0	2.7	2.70	
14	[0008154040]Java Script程序设计	3.0	专业课/限选课	考查	初修	91.00	3.0	4.1	12.30	

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序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
1	[0008200001]劳动教育（实践）	1.0	公共课/必修课	考查	初修	合格	1.0	1.5	1.50	
2	[0015000022]毛泽东思想和中国特色社会主义理论体系概论	3.0	公共课/必修课	考试	初修	71.00	3.0	2.1	6.30	
3	[0016000019]大学体育（四）	1.0	公共课/必修课	考试	初修	83.00	1.0	3.3	3.30	
4	[0017000TY9]大学英语（四）	1.5	公共课/必修课	考试	初修	85.00	1.5	3.5	5.25	
5	[15000084]形势与政策（四）	0.5	公共课/必修课	考查	初修	86.00	0.5	3.6	1.80	
6	[0008132002]操作系统	3.0	专业基础课/必修课	考试	初修	78.00	3.0	2.8	8.40	
7	[0008205041]软件工程概论	2.0	专业基础课/必修课	考查	初修	97.00	2.0	4.7	9.40	
8	[0008602013]概率论与数理统计	4.0	专业基础课/必修课	考试	初修	87.00	4.0	3.7	14.80	
9	[0008204002]Linux操作系统	3.0	专业课/必修课	考查	初修	71.00	3.0	2.1	6.30	
10	[0018002022]数据库系统概论	5.0	专业课/必修课	考试	初修	68.00	5.0	1.8	9.00	
11	[00001146]创新、发明与专利实务	2.0	公共课/任选课	考试	初修	100.00	2.0	5.0	10.00	
12	[00001194]经济与社会：如何用决策思维洞察生活	2.0	公共课/任选课	考试	初修	99.00	2.0	4.9	9.80	

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13	[00001304]婚恋-职场-人格	1.0	公共课/任选课	考试	初修	98.00	1.0	4.8	4.80	
14	[00001308]数字时代的智能技术	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
15	[00001310]智驭未来：AI工具辅助高效学习与科研	1.0	公共课/任选课	考试	初修	99.00	1.0	4.9	4.90	
16	[00001312]人工智能与创新	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
17	[00001314]科幻中的物理学	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
18	[0000T02]形象管理	2.0	公共课/任选课	考试	初修	100.00	2.0	5.0	10.00	
19	[0000ZB11]追寻幸福：中国伦理史视角	2.0	公共课/任选课	考试	初修	100.00	2.0	5.0	10.00	
20	[0000ZC20]音乐鉴赏	2.0	公共课/任选课	考试	初修	99.00	2.0	4.9	9.80	
21	[002110511G]豫西地区非遗文化概览	2.0	公共课/任选课	考查	初修	82.00	2.0	3.2	6.40	
22	[12001062]脑洞心理学：解锁生活中的奇妙心理密码	2.0	公共课/任选课	考查	初修	88.00	2.0	3.8	7.60	
23	[0008157061]Python程序设计	3.0	专业课/任选课	考查	初修	88.00	3.0	3.8	11.40	

学年学期：2025-2026学年第一学期

序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
1	[15000020]习近平新时代中国特色社会主义思想概论	3.0	公共课/必修课	考试	初修	81.00	3.0	3.1	9.30	
2	[24000001]就业指导	1.0	公共课/必修课	考查	初修	90.00	1.0	4.0	4.00	
3	[0008112051]软件需求分析	2.5	专业基础课/必修课	考查	初修	92.00	2.5	4.2	10.50	
4	[0008136042]软件工程导论课程设计	1.0	专业基础课/必修课	考查	初修	97.00	1.0	4.7	4.70	
5	[0008143061]软件系统建模与UML	3.0	专业基础课/必修课	考查	初修	83.00	3.0	3.3	9.90	

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6	[0008303031]计算机网络	3.5	专业基础课/必修课	考试	初修	78.00	3.5	2.8	9.80	
7	[0008133003]Java Web开发	3.0	专业课/限选课	考查	初修	92.00	3.0	4.2	12.60	
8	[0015001002]党史	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
9	[0015001003]新中国史	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
10	[0015001015]改革开放史	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
11	[0015001016]社会主义发展史	1.0	公共课/任选课	考试	初修	100.00	1.0	5.0	5.00	
12	[0008500401]算法设计与分析	3.0	专业课/任选课	考查	初修	93.00	3.0	4.3	12.90	

学年学期：2025-2026学年第二学期

序号	课程/环节	学分	类别	考核方式	修读性质	成绩	取得学分	绩点	学分绩点	备注
1	[0008022001]设计模式	2.5	专业课/必修课	考查	初修	91.00	2.5	4.1	10.25	
2	[0008134004]移动Web开发	3.0	专业课/必修课	考查	初修	100.00	3.0	5.0	15.00	
3	[0008208001]JAVA项目实训开发	3.0	专业课/限选课	考查	初修	93.00	3.0	4.3	12.90	
4	[00000001]人工智能综合能力提升培训	2.0	公共课/任选课	考查	初修	合格	2.0	1.5	3.00	
5	[0008024001]软件工程应用项目实践	2.0	专业课/任选课	考查	初修	89.00	2.0	3.9	7.80	